



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

Pattern Recognition [CSE-3274]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast single linkage and complete linkage clustering. (5)
- A)
- B) Analyse the concept of Convolutional Neural Network for the purpose of digit recognition. Describe the layers and justify their selection in terms of feature extraction and classification. (5)
- 2) Make use of bagging and boosting in the context of ensemble learning. What is the key difference between the two? (5)
- A)
- B) Analyse the role of learning rate and activation functions in the training process of a CNN. (5)
- 3) Summarize the steps involved in the Fuzzy C-Means algorithm with a suitable diagram. (3)
- A)
- B) Construct the formula of precision and recall, commonly used performance metrics for classifier evaluation. When would you prefer one over the other? (2)
- C) Estimate the final clusters for the given table data using single linkage and complete linkage clustering. Show the all calculation steps for the both clustering techniques. (5)
- | | A | B | C | D | E |
|---|---|---|---|---|---|
| A | 0 | 4 | 7 | 6 | 9 |
| B | 4 | 0 | 6 | 3 | 8 |
| C | 7 | 6 | 0 | 2 | 5 |
| D | 6 | 3 | 2 | 0 | 4 |
| E | 9 | 8 | 5 | 4 | 0 |
- 4) Identify the likelihood function in the context of Bayes classification? How is it mathematically defined for a given class and observation? (5)
- A)

B) Analyse the concept of Analysis of Variance (ANOVA) and its role in statistical pattern recognition. Describe how ANOVA helps in feature selection by comparing variances between classes. (5)

5) Identify Accuracy, Precision, Recall, and F1-Score for the given confusion matrix,

A)

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	35

(5)

B) Compare single layer and multi-layer perceptron using a suitable example. (3)

C) Illustrate how to select the value of K in K-means clustering? (2)

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